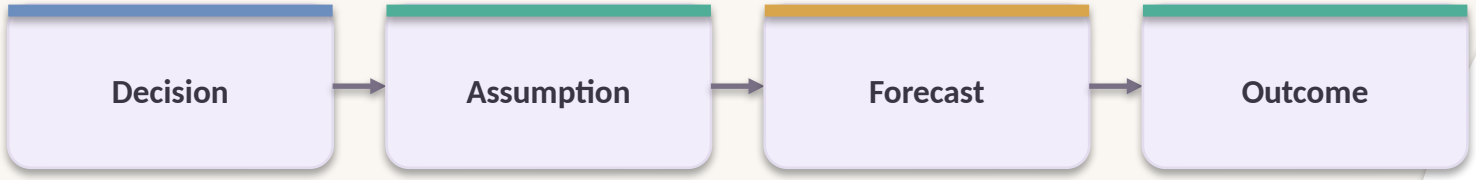
B.TECH ARTIFICIAL INTELLIGENCE & MACHINE LEARNING · CA2 CAPSTONE PROJECT

Decision Debt OS

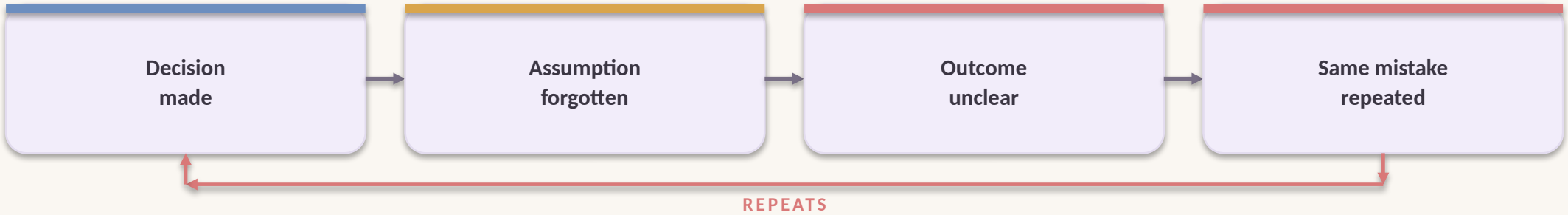
Turning business decisions into measurable learning



Name: Utsav Tyagi **PRN:** 26070126233 **Branch:** AI/ML

Division: C-3 **Institution:** Symbiosis Institute of Technology, Pune **Academic Year:** 2026-27

Decisions get made. Assumptions get forgotten.



Dashboards and BI tools show what happened. Project trackers show what is scheduled. Neither one records **what a team believed would happen** before they acted — so when the belief turns out to be wrong, nothing captures that it was ever tested, and the same unexamined assumption drives the next decision.

Four Recent Papers, Four Different Gaps

AUTHOR(S) & YEAR	TITLE	PROS	CONS	DIFFERENTIATION FROM DECISION DEBT OS
Scheibel & Rinderle-Ma (CAiSE 2023)	An End-to-End Approach for Online Decision Mining and Decision Drift Analysis	Monitors decision rules at runtime, not just after the fact; detects “decision drift”; validated on 4 synthetic + 1 real-life log	Tracks how an automated decision RULE changes — not whether a human's stated assumption about an outcome came true; needs existing event logs	We track a human assumption vs. a measured outcome; they track automated rule drift inside a process log — adjacent problem, different object
Goossens, De Smedt & Vanthienen (2024, KU Leuven)	Extracting Process-Aware Decision Models from Object-Centric Process Data	First object-centric decision-mining algorithm (IODDA); discovers decision structure + rules directly from logs; open benchmark released	Purely retrospective — reconstructs a model from data that already exists; can't recover an assumption that was never logged; no deadline/validation concept	We capture the assumption BEFORE the outcome is known (forward-looking); they mine structure AFTER the fact (backward-looking) — complementary, not a substitute
Natarajan, Mathur, Sidheekh, Stammer & Kersting (AAAI-25)	Human-in-the-loop or AI-in-the-loop? Automate or Collaborate?	Directly tackles when to automate a decision vs. keep a human in control; argues evaluation over-weights the model, under-weights the human expert	A conceptual/positional framework, not a deployable system; no mechanism to record or later validate the human's stated assumption	We operationalise exactly this division of labour: the model scores risk, a human approves + states the assumption, the system checks the outcome later
Gruetzemacher, Lee & Paradice (2024)	Calibration Training for Improving Probabilistic Judgments Using an Interactive App	Shows forecast calibration can be measurably improved with short, repeated feedback — supports asking for a stated confidence number	Small exploratory study (153 students, a sports-forecasting tournament); trains general forecasting skill, not tied to any business decision or metric	We don't train forecasting skill in the abstract — we capture a one-off, decision-specific forecast and check it against a real business metric

All four papers published 2023–2025 (peer-reviewed / top-tier preprint) — verified titles, authors and venues. See References for full citations.

CORE INSIGHT

A decision is not finished when it is made.

> *"A business decision is incomplete until its original assumption is compared with the real-world outcome."*

This gap has a name:

Decision Debt — the accumulation of decisions whose assumptions were never checked against reality.

What Is Decision Debt?

Unresolved Assumptions

A decision was made, but its underlying belief was never written down in a checkable form.

Overdue Reviews

A deadline for checking the outcome has passed, and no one has looked at the result.

Invalidated Assumptions Still Driving Work

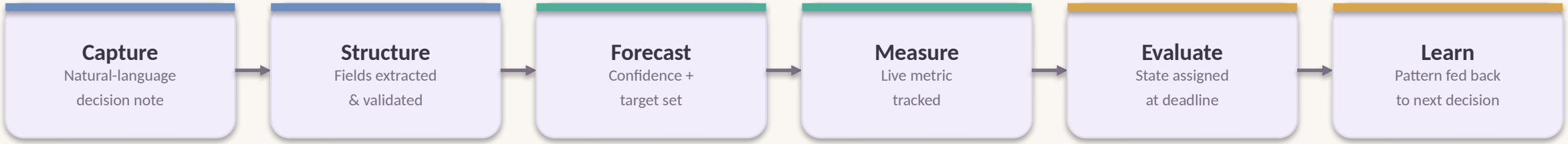
The assumption was checked, found false, but the decision or policy it justified is still active.

Repeated Forecasting Errors

The same type of assumption keeps failing across different decisions, and no one notices the pattern.

THE SOLUTION

Capture → Structure → Forecast → Measure → Evaluate → Learn



UNSTRUCTURED INPUT

"Let's offer the 10% discount to high-risk customers — I think it'll cut churn to around 18% within a month."

STRUCTURED DECISION RECORD

INTERVENTION 10% retention discount

TARGET METRIC 30-day churn, high-risk cohort

FORECAST 25% → 18%

CONFIDENCE 62%

DEADLINE +30 days

WORKED EXAMPLE

One Decision, Fully Specified

> *"We believe a 10% retention discount will reduce 30-day churn among high-risk customers from 25% to 18%."*

OWNER

Retention Lead, Growth Team

CONFIDENCE (FORECAST)

62% self-reported probability

INTERVENTION

10% discount, auto-applied at checkout

TARGET METRIC

30-day churn rate, high-risk cohort

DEADLINE

30 days from decision date

ESTIMATED COST

~$4.80 / customer (10% of avg. order)

KILL CONDITION

If interim churn > 25% at day 15, stop offer

ACTUAL RESULT

Pending — 30-day window not yet elapsed

Baseline shown here (25%) is one risk-tier definition; the dataset's observed high-risk cohort baseline is 37% — see Slide 7.

From Customer Record to Decision Evaluation

Customer profile → Churn-risk prediction → Retention decision → Treatment / control group → 30-day outcome → Decision evaluation

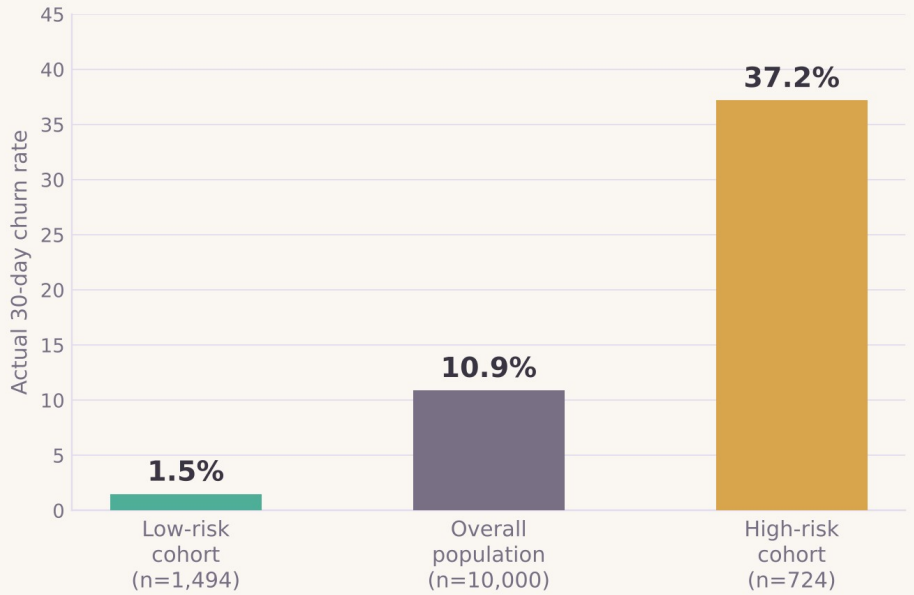
SYNTHETIC CUSTOMER-RETENTION DATASET

RECORDS 10,000 fictional customer rows

FIELDS age, gender, country, income, spending score, purchases, avg. order value, membership years, site visits/mo, cart-abandon rate, churn flag, feedback text, last purchase date

STATUS Clearly labelled synthetic data — used for demonstration only; not a real company's data

Real, observed churn rate by risk segment



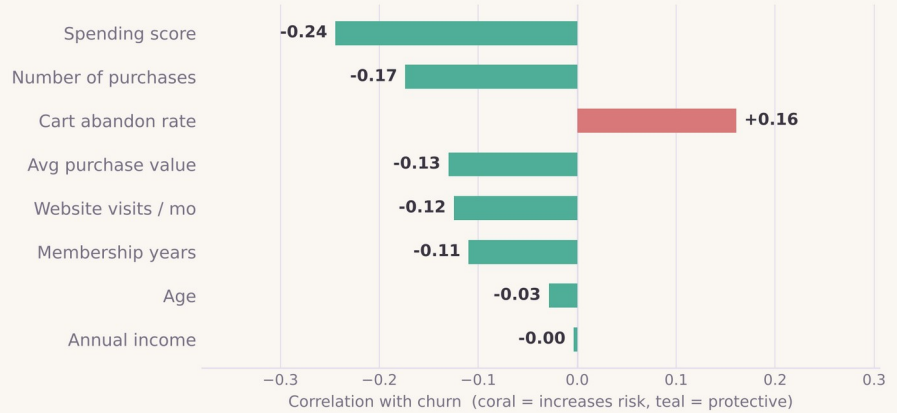
DATASET PREVIEW

What the Raw Rows Actually Look Like

First 8 rows of Customer_Transactions.csv
(8 of 14 columns shown)

ID	Age	Gender	Country	Income	Spend Score	Cart Abandon	Churned
1	37	Male	Germany	85,886	14	0.95	0
2	40	Male	India	41,041	4	0.21	0
3	69	Female	Australia	143,869	59	0.08	0
4	30	Male	UK	87,261	45	0.61	0
5	69	Female	UK	110,678	40	0.49	0
6	28	Male	Australia	90,330	2	0.81	1
7	40	Male	India	27,302	51	0.8	0
8	21	Male	Brazil	39,171	83	0.91	0

Which features actually move churn
(real, signed correlations)



10,000 rows × 14 columns

Clearly labelled synthetic / fictional data, generated for this project — not drawn from any real company's customers. No column was dropped or reshaped for this preview; these are the literal supplied values.

Cart-abandon rate is the only feature that **increases** churn risk among these eight — everything else (spending, purchase frequency, tenure) is protective. This matches the model's own feature-importance ranking one slide over.

MACHINE LEARNING COMPONENT

Predicting Risk, Not Proving Causation

HOW IT'S BUILT

TASK Binary classification — churned vs. retained

FEATURES Spending score, cart-abandon rate, site visits/mo, purchases, avg. order value, membership years, income, age, gender, country

PREPROCESSING Standardization (numeric) + one-hot encoding (categorical)

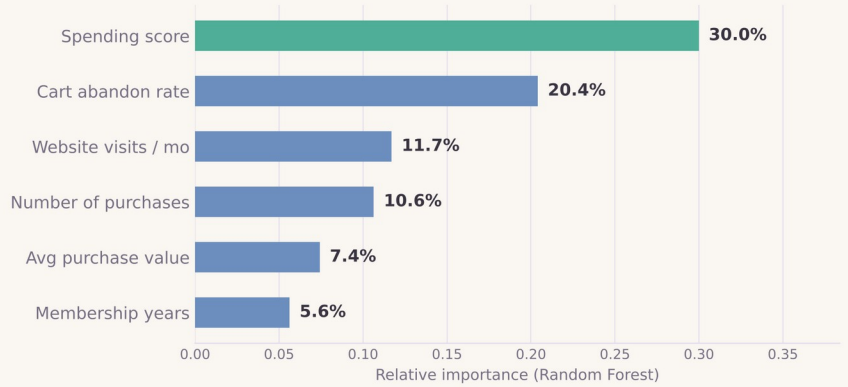
SPLIT 80 / 20 train-test, stratified by class

MODELS COMPARED Logistic Regression vs. Random Forest (300 trees)

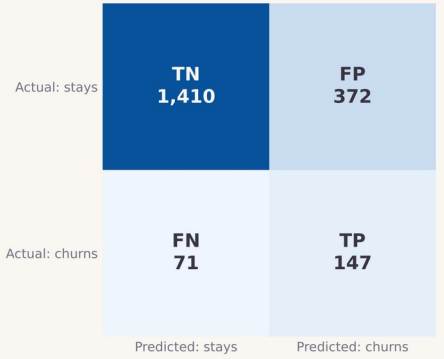
0.28	0.67	0.40	0.84
Precision	Recall	F1-score	ROC-AUC

Random Forest · held-out test set · n = 2,000 (live, reproducible — same numbers the running Verdikt prototype shows)

Top churn-risk drivers



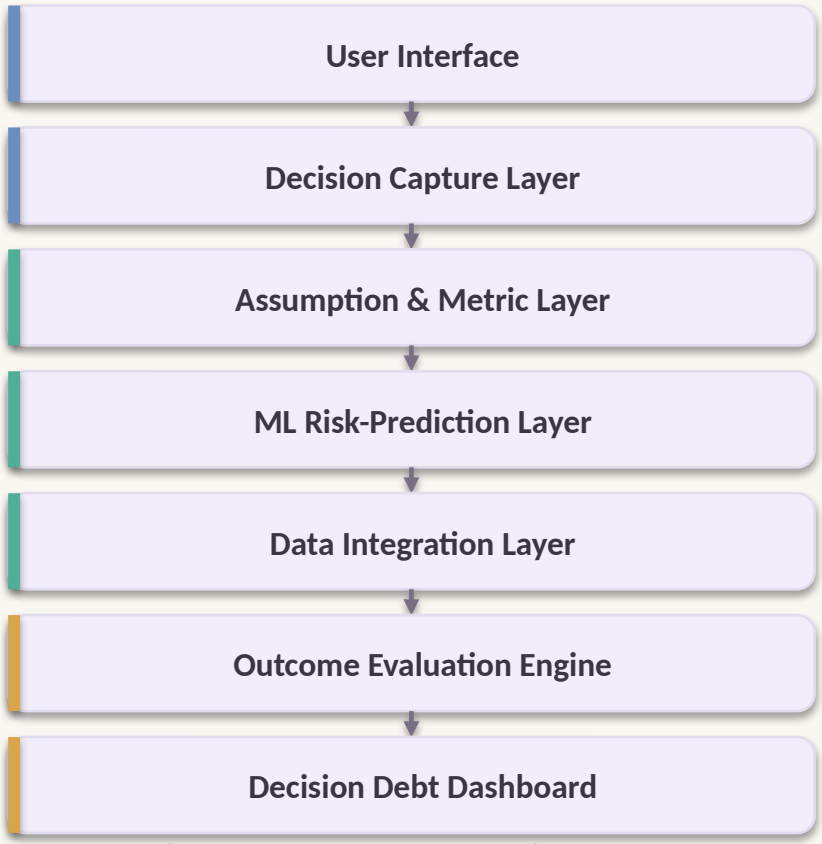
Confusion matrix



A model that flags churn risk does not prove an intervention caused a customer to stay — prediction identifies candidates for a decision; it does not validate the decision's outcome.

SYSTEM ARCHITECTURE

Seven Layers, One Feedback Loop



Where a manager types a decision in plain language

Parses the note into a draft structured record

Binds the decision to one measurable target metric

Scores churn risk to prioritise which customers/decisions matter

Pulls in product, billing and CRM data to track the metric live

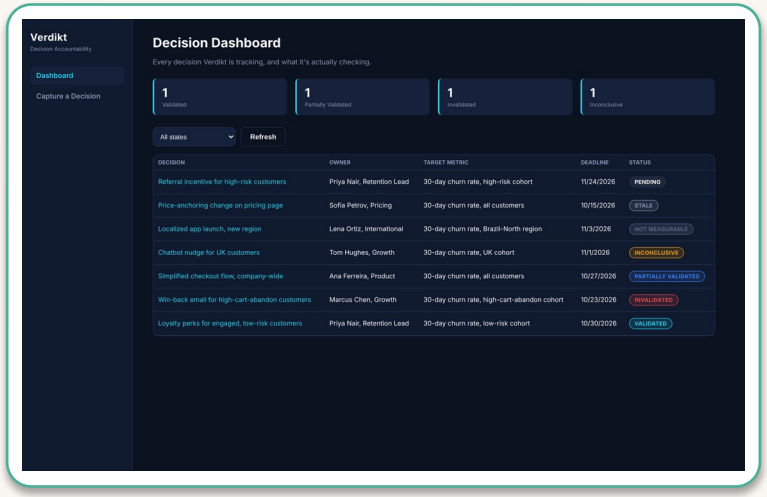
Checks deadline, sample size and target at review time

Shows every decision's state — and what's overdue or invalidated

BUILT WITH (REAL, RUNNING PROTOTYPE)

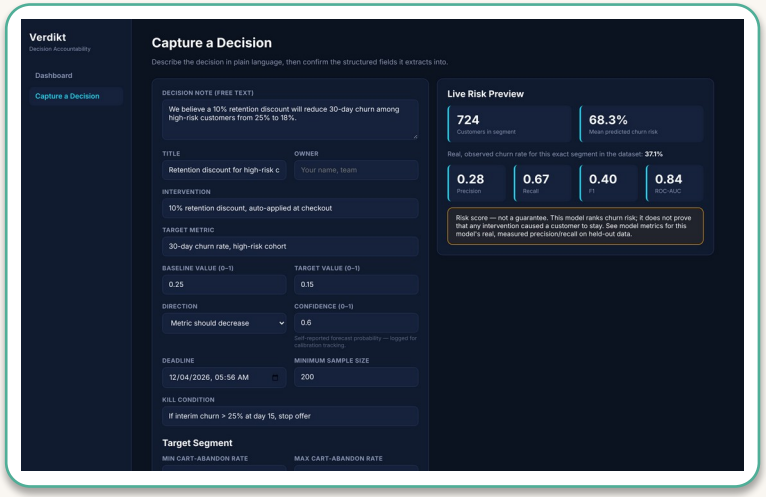
Python | FastAPI | SQLAlchemy | Scikit-learn | React + TypeScript

Not a Mockup. A Running Application.



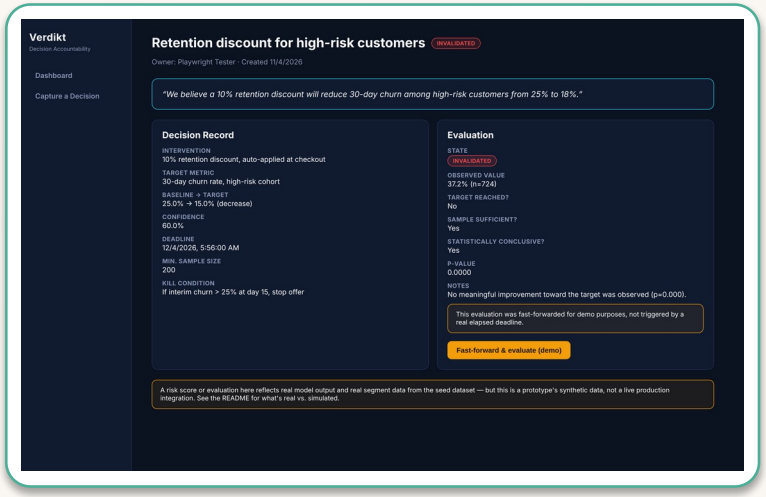
Decision Dashboard

7 seeded decisions, every validation state represented



Capture — Live Risk Preview

Real segment scoring as the form is filled in



Decision Detail — Evaluated

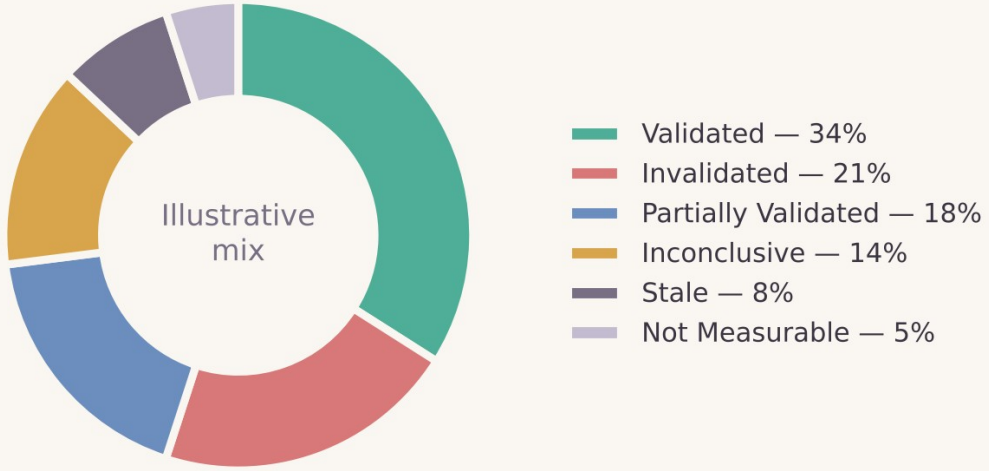
Real p-value, real state: Invalidated

All three screens captured from the actual running app (FastAPI backend + React frontend, real SQLite data) — not a design mockup.

Four Checks, Six Possible States

- Was the target metric reached?
- Has the review deadline passed?
- Was the sample size large enough to trust the result?
- Is the result statistically conclusive, or still noisy?

Illustrative state mix — simulated for demonstration



Validated / Invalidated = target clearly met or missed, with enough data. Partially Validated = directionally right, underpowered. Inconclusive = evidence too weak either way. Stale = deadline passed, never reviewed. Not Measurable = metric or data unavailable.

Honest Positioning: The Pieces Exist. The Workflow Is New.

	Captures decisions	Explicit assumption	Forecasts outcome	Closes loop w/ measured outcome	Built for product & retention teams
BI dashboards (Looker, Tableau, Power BI)	–	–	–	–	–
Jira / Linear / Notion	◐	–	–	–	◐
Corporate prediction markets (P0, Hypermind)	–	◐	✓	◐	–
Investment decision trackers	✓	◐	◐	◐	–
Enterprise decision intelligence (Vigil, org-memory patents)	✓	✓	◐	◐	–
Prediction/outcome journals (RationalGo, Reckon, Condelo)	✓	◐	✓	◐	–
Decision Debt OS (this project)	✓	✓	✓	✓	✓

NOVELTY, SCORED HONESTLY

3/10
Algorithmic & technical novelty

8/10
Workflow & combination novelty

Not a new machine-learning technique. The real argument is the workflow combination, and even that is close to at least three small existing tools.

B2B SaaS, 20–150 Employees, Active Growth Teams

BUYER

VP Product or Head of Growth / Retention at a venture-backed B2B SaaS company

INITIAL USE CASE

Churn-intervention decisions — discounts, win-back campaigns, onboarding changes

PILOT PRICE

$499 / month per workspace

EXPANSION PATH

Retention decisions → pricing & packaging → feature launches → org-wide decision registry

BUSINESS MODEL

B2B SaaS subscription, per-workspace, usage-based tiers above the pilot

Business hypothesis — not yet validated with paying customers or signed pilots.

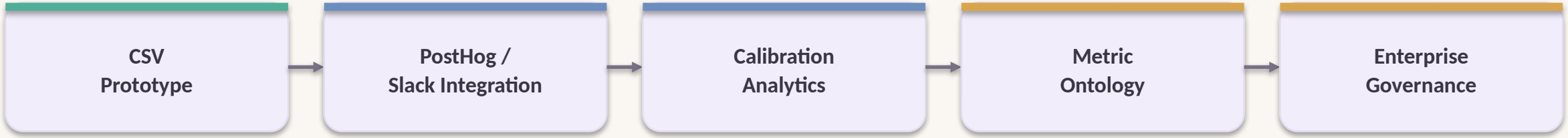
What Could Go Wrong — And the Mitigation

RISK	MITIGATION
Synthetic data cannot prove real-world value	Pilot with design-partner companies' real (anonymised) data before any efficacy claim.
No guaranteed causal inference from churn prediction	Treat scores as risk triage only; require a held-out comparison group before crediting any intervention.
Metric-binding errors — wrong metric tied to a decision	Shared metric ontology plus reviewer sign-off before a decision record goes live.
Poor data quality / integration gaps	Data-quality checks at ingestion; low-confidence source feeds are flagged, not trusted silently.
Forecast manipulation — gaming confidence scores	Track forecaster calibration over time so gaming becomes visible, not just the raw score.
Privacy — customer PII inside decision records	Field-level encryption, role-based access, minimal PII in ML features.
False alerts / alert fatigue from risk scoring	Tunable thresholds and a batched digest instead of per-customer pings.
User resistance / adoption friction	Natural-language capture to avoid new process overhead; start with one team, not the whole org.
Prior-art & patent overlap (e.g., EP1402435A4, US20250217680A1)	Freedom-to-operate review before commercial launch; keep claims to the workflow, not the method.

Decision Debt OS Doesn't Replace Judgment. It Makes It Checkable.

> *"Decision Debt OS does not replace human judgment. It makes the assumptions behind human decisions visible, measurable and reviewable."*

ROADMAP



"The goal is not to automate responsibility. The goal is to prevent important decisions from disappearing into memory."

Related Work & Sources

LITERATURE REVIEW — LAST 3 YEARS

1. Scheibel, B. & Rinderle-Ma, S. (2023). An End-to-End Approach for Online Decision Mining and Decision Drift Analysis in Process-Aware Information Systems. *CAiSE 2023 Forum / arXiv:2303.03961*

2. Goossens, A., De Smedt, J. & Vanthienen, J. (2024). Extracting Process-Aware Decision Models from Object-Centric Process Data. *arXiv:2401.14847 (KU Leuven)*

3. Natarajan, S., Mathur, S., Sidheekh, S., Stammer, W. & Kersting, K. (2025). Human-in-the-loop or AI-in-the-loop? Automate or Collaborate? *Proc. AAAI-25, 39(27), 28594–28600*

4. Gruetzemacher, R., Lee, K.B. & Paradice, D. (2024). Calibration Training for Improving Probabilistic Judgments Using an Interactive App. *Futures & Foresight Science, 6(2), e177*

BACKGROUND & PRIOR ART

1. Organisational Memory and Decision Rationale — ScienceDirect

2. Operational Decision Discovery from Data — Springer

3. Corporate Prediction Markets & Decisions — Cambridge / Experimental Economics

4. Incentive-Compatible Forecasting — INFORMS / Management Science

5. P0 — Internal Prediction Markets — priorityzero.ai

6. Hypermind — Crowd Forecasting — hypermind.com

7. Vigil — Decision Tracking Use Cases — trustvigil.com

8. Patent EP1402435A4 — Google Patents

9. Patent US20250217680A1 — Google Patents